Option 1: Parking Lot Management Application

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Approach and Design

```
        String[] smallSpots = new String[n];
        String[] largeSpots = new String[n];
        String[] oversizedSpots = new String[n];
```

Approach and Design Continued

```java
        if (command.equals("Enter")) {
            String size = normalizeSize(parts[1]);
            String plate = parts[2];

            String[] spots = getSpots(size, smallSpots, largeSpots, oversizedSpots);
            handleEnter(size, plate, spots);
        } else if (command.equals("Leave")) {
            String size = normalizeSize(parts[1]);
            String plate = parts[2];

            String[] spots = getSpots(size, smallSpots, largeSpots, oversizedSpots);
            handleLeave(size, plate, spots);
        } else if (command.equals("Status")) {
            printStatus(smallSpots, largeSpots, oversizedSpots);
        }
```

Key Files and Folders

```
README.md     run_tests.sh src          tests

./src:
Parking.class Parking.java

./tests:
actual1.txt   actual2.txt   actual3.txt   expected1.txt expected2.txt expected3.txt input1.txt    input2.txt    input3.txt
```

Process to Run, Test, and Verify

```
# 2026ParkingLotManagementApplication

# compile
javac src/Parking.java

# run test case
java -cp src Parking tests/input1.txt > tests/actual1.txt

# verify test case
diff tests/expected1.txt tests/actual1.txt

# run all test cases
./run_tests.sh
```

Process to Run, Test, and Verify Continued

```bash
#!/bin/bash

javac src/Parking.java || exit 1

for test in tests/input*.txt
do
    base=$(basename "$test" .txt)
    number=${base#input}

    expected="tests/expected${number}.txt"
    actual="tests/actual${number}.txt"

    java -cp src Parking "$test" > "$actual"

    if diff -q "$actual" "$expected" > /dev/null; then
        echo "Test ${number} passed"
    else
        echo "Test ${number} failed"
        diff "$expected" "$actual"
    fi
done
```

Test Data

```
2
Enter Small 1111
Enter Small 2222
Enter Small 3333
Enter Large 4444
Enter Large 5555
Enter Large 6666
Enter Oversized 7777
Enter Oversized 8888
Enter Oversized 9999
Status
Leave Small 1111
Enter Small 3333
Status
Exit
```

```
small vehicle with license plate 1111 is parking in spot 1.
small vehicle with license plate 2222 is parking in spot 2.
No more small spots available for vehicle with id 3333.
large vehicle with license plate 4444 is parking in spot 1.
large vehicle with license plate 5555 is parking in spot 2.
No more large spots available for vehicle with id 6666.
oversized vehicle with license plate 7777 is parking in spot 1.
oversized vehicle with license plate 8888 is parking in spot 2.
No more oversized spots available for vehicle with id 9999.
Status:
Small spots:
1 1111
2 2222
Large spots:
1 4444
2 5555
Oversized spots:
1 7777
2 8888
small vehicle with license plate 1111 is leaving spot 1.
small vehicle with license plate 3333 is parking in spot 1.
Status:
Small spots:
1 3333
2 2222
Large spots:
1 4444
2 5555
Oversized spots:
1 7777
2 8888
```